

ANKUSH SAROJ

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering student specializing in Embedded Systems, Firmware Development, and IoT. Skilled in Embedded C, C++, STM32, ESP32, FreeRTOS, UART, SPI, and I2C communication protocols. Experienced in microcontroller programming, hardware-software integration, real-time embedded applications, and IoT system development. Passionate about designing reliable, high-performance embedded solutions and seeking opportunities as an Embedded Firmware Engineer.

EXPERIENCE

Embedded System and IoT Intern

01/2026 – 03/2026

Maincrafts Technology 🌐

Uttar Pradesh

- Developed and optimized Embedded C firmware for microcontroller-based systems involving sensor interfacing, actuator control, and real-time data acquisition.
- Integrated UART, SPI, I2C, MQTT, HTTP, and TCP/IP protocols to enable reliable device communication and cloud connectivity.
- Conducted hardware debugging, firmware validation, and system-level testing to improve system reliability and performance

Research Intern - 6G Wireless Communication and Localization, NIT Patna

05/2025 – 07/2025

NIT Patna 🌐

NIT Patna

Completed a research internship on 6G wireless communication and localization.

- Explored URLLC+ for enabling real-time autonomous systems.
- Researched high-precision localization and sensor fusion for sub-meter accuracy.
- Analyzed 6G enablers like terahertz channels, massive MIMO and AI-driven signal optimization.
- Designed drone-based localization for smart agriculture using GPS and image processing.

EDUCATION

B.Tech in Electronics and Communication Engineering

2023 – Present

Central University of Karnataka 🌐

Kalaburagi

CGPA: 8.22/10 (till 6th Semester)

10+2 with Physics, Chemistry, Mathematics and Computer Science

2019 – 2022

Jawahar Navodaya Vidyalaya Narayanpur Pratapgarh 🌐

Pratapgarh, India

PROJECTS

AI-Powered IoT-Based Smart Health Monitoring and Diagnostic System (NEXA) 🌐

01/2026 – 03/2026

Real-Time Vital Signs Monitoring with AI-Based Analysis, Voice Interface, and Cloud Integration

Developed an ESP32-based AI-powered IoT health monitoring system integrating heart rate, SpO₂, temperature and GPS sensors for real-time patient monitoring. Implemented AI-based health analysis, STT/TTS voice interaction, cloud dashboard visualization, and Telegram API-based emergency alerts with live location sharing for remote healthcare monitoring.

Automated-Waste-Segregation-System-for-Efficient-Recycling 🌐

03/2025 – 03/2025

Developed an Arduino-based smart waste segregation system that automatically classifies dry, wet, and metallic waste using sensors and automated sorting mechanisms, improving recycling efficiency and reducing manual effort.

RFID-Based Attendance System

09/2025 – 09/2025

Central University of Karnataka

Developed an RFID-based smart attendance system using Arduino and the RC522 RFID module to automate attendance recording. Integrated LCD display for real-time user feedback and implemented automatic Excel-based data logging, enhancing record accuracy and eliminating manual attendance management.

Smart Home Automation using ESP32 & Blynk

02/2025 – 03/2025

Central University of Karnataka

Built an IoT-enabled Smart Home Automation system using ESP32 and Blynk, enabling real-time remote appliance control via smartphone, showcasing strong expertise in embedded systems, Wi-Fi communication, and hardware–cloud integration.

AI Voice Assistant using ESP32 (STT + LLM + TTS)

10/2025 – 11/2025

Central University of Karnataka

Developed a WiFi-enabled AI voice assistant integrating Deepgram Speech-to-Text, Groq LLaMA 3.1 LLM, and Deepgram Text-to-Speech to process voice queries and deliver intelligent spoken responses via DF Mini Player and speaker.

SKILLS

Programming Language

C, C++, Python, Embedded C, MicroPython, 8051 Microcontroller Programming, HTML, CSS, JavaScript

Tools & Development Platforms

Visual Studio Code, Code::Blocks, Jupyter Notebook, Xilinx Vivado, Arduino IDE, STM32CubeIDE, KiCad, EDA Tools

Core Competencies

AWS Fundamentals, Image Processing, VLSI & Front-End Design, Data Structures & Algorithms (C++), Drone Hardware Integration & Calibration, Soldering

Embedded Systems & Hardware

Arduino Uno, ESP32, STM32, Raspberry Pi, Microcontroller Programming, RTOS (FreeRTOS), FPGA Programming, PCB Design, Embedded Linux, GDB Debugging

Communication Protocols

UART, USART, SPI, I2C, MQTT, HTTP, TCP/IP, CAN

CERTIFICATION

Embedded System using.

Arduino

GeeksforGeeks

Drone Hardware Integration and Calibration

NIT Patna

Summer Internship on 6G

Wireless Communication and Localization

Front - End VLSI Design Using.

Verilog

Internshala

Embedded C

NIELIT Calicut

VLSI For Beginners

Nielit Calicut

Wireless Sensor Network (With IoT Application)

Meity

Embedded System & IoT Internship

LANGUAGES

Hindi

English

Telugu